



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., V & CSE

Course Code & Title: EC8691 & Microprocessor and Microcontroller

Name of the Faculty member (s): Dr.A.Azhagu Jaisudhan Pazhani

Innovative Practice Description

- **Unit / Topic: III / Case Studies** – Keyboard and LCD Display & alarm Controller
- **Course Outcome:** CO3: The students will be able interface 8086 microprocessor with supporting chips.
- **Topic Learning Outcome:** TLO9: The students will be able to implement the real time applications and write the assembly language program for real time applications
- **Activity Chosen:** Flipped class room

- **Justification:**

It allows students to learn at their own pace, encourages students to actively engage with lecture material, frees up actual class time for more productive, creative, and active learning activities, and students take charge of their own learning. After completing this assignment, I will be able to assess the students' comprehension levels.

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

Students were assigned a specific topic to research on their own. Resources such as a reference book and videos link were given. Students are expected to prepare in greater detail than in the past. Students are divided into groups and each group is assigned an assignment to do. Bring the class back together to share the presentation of work of each group with the rest of the class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2
CO3	2	3	3	1	1	2	1	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO9	PSO2
	2	2
Justification for correlation	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	It helps the students to identify and interface the suitable peripheral components for real time interface based applications.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students believe that their self-learning has improved. They learn how to work as a team and communicate with one another.

- ❖ **Benefit of the practice:**

Every student had an equal opportunity to participate in this activity. The activity's success was assessed by asking the same question in Internal Assessment Test II – Approximately 80% of students correctly replied.

- ❖ **Challenges faced in implementation:**

The fundamental issue is that few pupils have been exposed to flipped classrooms. Students who struggle with self-discipline may show up for class without having fully comprehended the material. To encourage all children to participate, I began by praising others, taking turns for equitable participation, and allowing students to make decisions together.

References:

- ❖ Yu-Cheng Liu, Glenn A.Gibson, “Microcomputer Systems: The 8086/ 8088 Family – Architecture, Programming and Design”, Second Edition, Prentice Hall of India 2007.
- ❖ Mohamed Ali Mazidi, Janice Gillispie Mazidi, Rolin Mckindlay, “The 8051 Microcontroller and Embedded systems: Using Assembly and C”, Second Edition Pearson Education, 2011.



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- **Unit / Topic: V – Waveform Generation using 8051**
- **Course Outcome:** CO5: The students will be able demonstrate hardware interfacing peripherals.
- **Topic Learning Outcome:** TL14: The students will be able to implement the real time applications and write the assembly language program for real time applications
- **Activity Chosen: Jigsaw**
- **Justification:**

Jigsaw is a collaborative group activity with a twist: students effectively teach each other (with the teacher's guidance). Students learn through the process of communicating with one another about a given skill or procedure, topic or problem. This topic and its related to waveform generation using 8051 microcontroller are more suitable for a Jig-Saw cooperative learning activity in which students get exposure to write programs using 8051 assembly languages. Students should be able to enhance their communication, programming knowledge and teamwork, and self-learning by this activity
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

The heterogeneous groups with six members were formed. Each group formed by bright students, average students, and slow learners, consists of 4 members. One group contains 5 members. Classification of students such as a bright student, average students, and slow learners is based on their academic performance. The total class strength is 41 (Boys and Girls). Seven heterogeneous groups were formed based on academic performances such as the CGPA of end semester examination. The table1 shows the name of the groups.

Sl.No	Group Name	Number of members
1	Mercury	6
2	Venus	6
3	Mars	6
4	Jupiter	6
5	Saturn	6
6	Neptune	6
7	Pluto	5

Sub Topics:

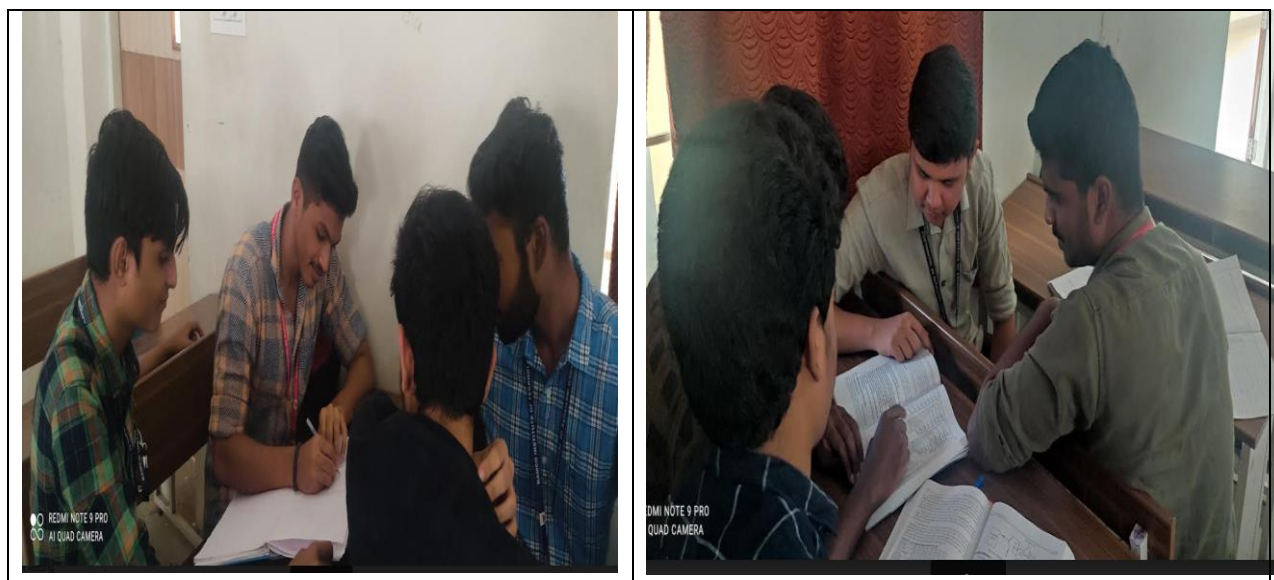
- Square Wave form Generations
- Triangle Wave form Generations
- Sawtooth Wave form Generations
- Non Uniform Pulse Wave form Generations
- Sine Wave form Generations

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO10	PO12	PSO2
CO5	2	3	3	1	1	2

• PO / PSO mapped:

Innovative practice	PO2	PSO2
	3	2
Justification for correlation	It helps the students to provide solution for complex problems such as generating sine, triangular waves from digital input signals using 8051 microcontroller. Hence it is mapped to level 3.	It helps the students to identify and interface the suitable peripheral components for real time interface based applications.

• Images / Screenshot of the practice:**• Reflective Critique:****❖ Feedback of practice from students and other stakeholders:**

It piqued the curiosity of the students. They learn how to work as a team and communicate with one another.

❖ **Benefit of the practice:**

Every student had an equal opportunity to participate in this activity. The success of the exercise was assessed by asking the same question in Internal Assessment Test III, which was answered correctly most of the students.

❖ **Challenges faced in implementation:**

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ Yu-Cheng Liu, Glenn A.Gibson, “Microcomputer Systems: The 8086/ 8088 Family – Architecture, Programming and Design”, Second Edition, Prentice Hall of India 2007.
- ❖ Mohamed Ali Mazidi, Janice Gillispie Mazidi, Rolin Mckindlay, “The 8051 Microcontroller and Embedded systems: Using Assembly and C”, Second Edition Pearson Education, 2011.

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INNOVATIVE TEACHING METHOD

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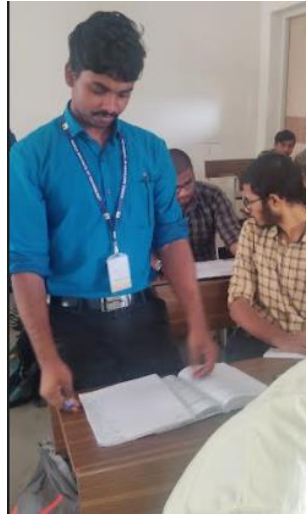
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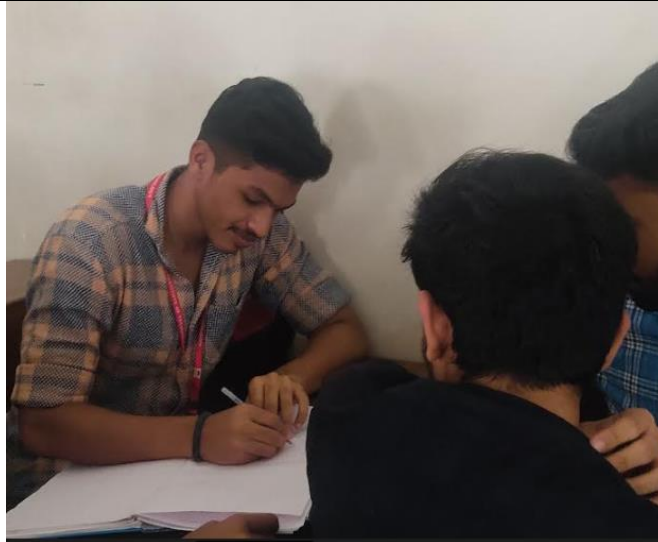
Innovative Teaching Method Execution

UNIT I – THE 8086 MICROPROCESSORS

Assembly Language Programming – Mind Map on 02.09.2021



Byte String Manipulation – One Minute Paper on 09.09.2021



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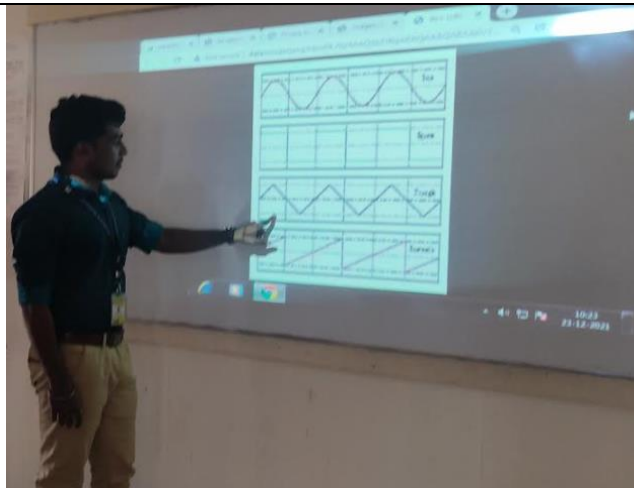
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UNIT II –8086 SYSTEM BUS STRUCTURES

System Design using 8086 - Z to A Approach on 30.09.2021



Advanced Processors – Presentation on 08.10.2021



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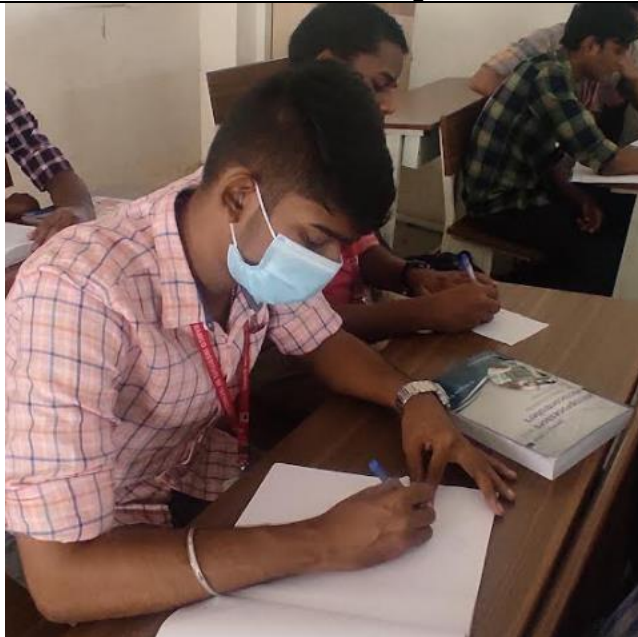
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UNIT III – I/O Interfacing

Memory Interfacing – Zero Minute Speech on 09.10.2021



DMI Controllers – Exit Slip on 25.10.2021



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UNIT IV – MICROCONTROLLERS

8051 Architecture – Mind Map on 28.10.2021



Instruction Set – Open Book Test on 09.11.2021

Open Book Test: Instruction set: Tulageswaran

1. Data transfer instructions
2. Arithmetic instructions
3. Logical Instructions
4. Branching Instructions
5. Boolean Instructions

These are the five types
of Instruction set.

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UNIT V – INTERFACING MICROCONTROLLERS

Interrupt Programming – Role Play on 21.11.2021



Microprocessors and Microcontrollers Comparison – One Minute Paper on 23.11.2021

